CarDekho Used Car Dataset

Exploratory Data Analysis Report

Indian Automotive Market | Price Prediction Foundation

Tools: Python · Pandas · Seaborn · Matplotlib · SciPy

Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the CarDekho Used Car Dataset, encompassing over 15,000 used car listings from India's dynamic secondary automobile market. The analysis investigates 13 key business questions spanning brand distribution, pricing dynamics, engine specifications, fuel preferences, transmission patterns, and seller channel behavior.

The Indian used car market exhibits a pronounced dual-segment structure: a high-volume mass-market tier dominated by Maruti, Hyundai, and Honda, and a low-volume luxury tier anchored by Ferrari, Rolls-Royce, and Bentley. Key pricing drivers are engine capacity and maximum power output, while depreciation factors such as vehicle age and kilometers driven show surprisingly weak correlations with resale value.

These findings establish the analytical foundation for building a robust machine learning price prediction model. Critical preprocessing steps — including log transformation of selling price, outlier treatment for luxury brands, and feature selection to address engine-power multicollinearity — are recommended before model training.

15,000+	12	13	0.75
Used Car Listings	Dataset Features	EDA Questions	Max Power–Price Correlation

Dataset Overview

About the Dataset

The CarDekho Used Car Dataset captures real-world listings from India's largest used car marketplace. It includes a broad spectrum of vehicles ranging from budget hatchbacks to ultra-luxury supercars, providing rich material for both market analysis and machine learning applications.

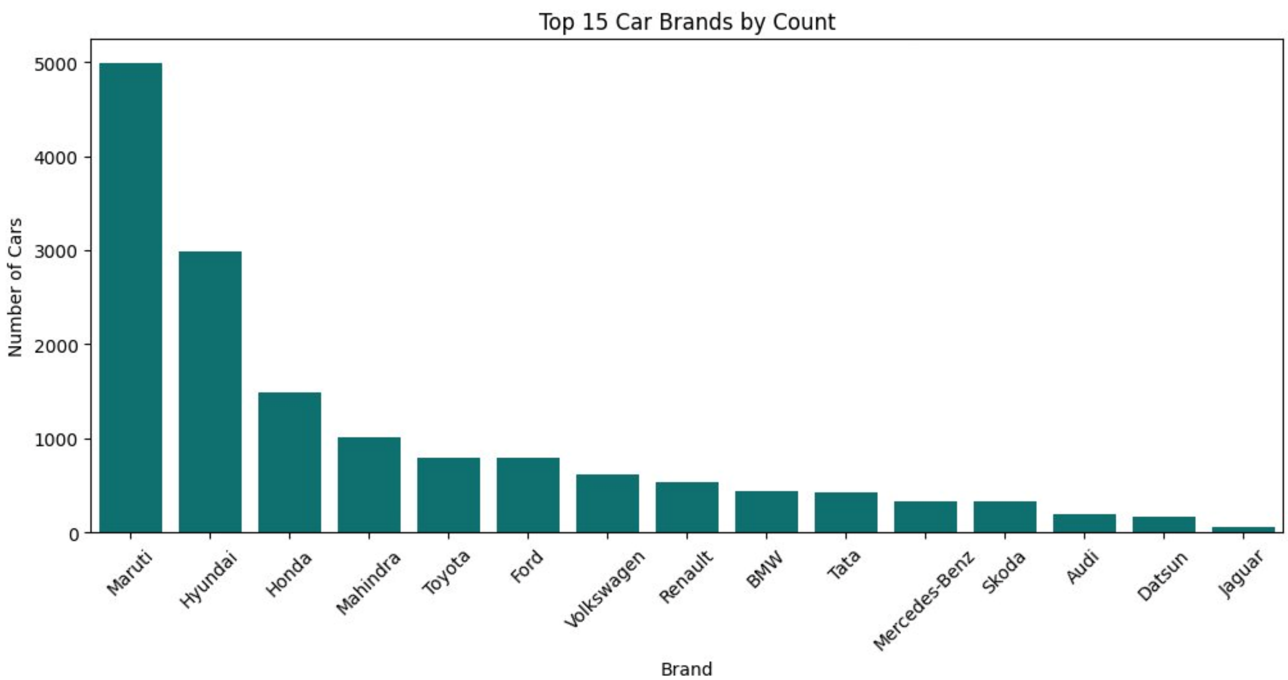
Property	Details
Source	CarDekho Used Car Marketplace
Domain	Automotive / Used Car Market — India
Primary Use Case	Used Car Price Prediction (Supervised Regression)
Format	CSV
Total Features	12 (7 Numerical, 5 Categorical)
Target Variable	selling_price (INR)

Feature Reference

#	Feature	Type	Description
1	**car_name**	Categorical	Full name of the car including brand and specific model
2	**brand**	Categorical	Brand name of the car
3	**model**	Categorical	Exact model name of the car under a particular brand
4	**seller_type**	Categorical	Type of seller (Individual / Dealer / Trustmark Dealer)
5	**fuel_type**	Categorical	Fuel type (Petrol / Diesel / CNG / LPG / Electric)
6	**transmission_type**	Categorical	Transmission type (Manual / Automatic)
7	**vehicle_age**	Numerical	Number of years since the car was originally purchased
8	**mileage**	Numerical	Fuel efficiency in km per litre (kmpl)
9	**engine**	Numerical	Engine displacement capacity in CC (cubic centimeters)
10	**max_power**	Numerical	Maximum power output in BHP
11	**seats**	Numerical	Total seating capacity in the car
12	**selling_price**	Numerical	Target variable — listed sale price on the platform (INR)

Exploratory Data Analysis

Q1 What are the top 15 most common car brands?



Key Insights

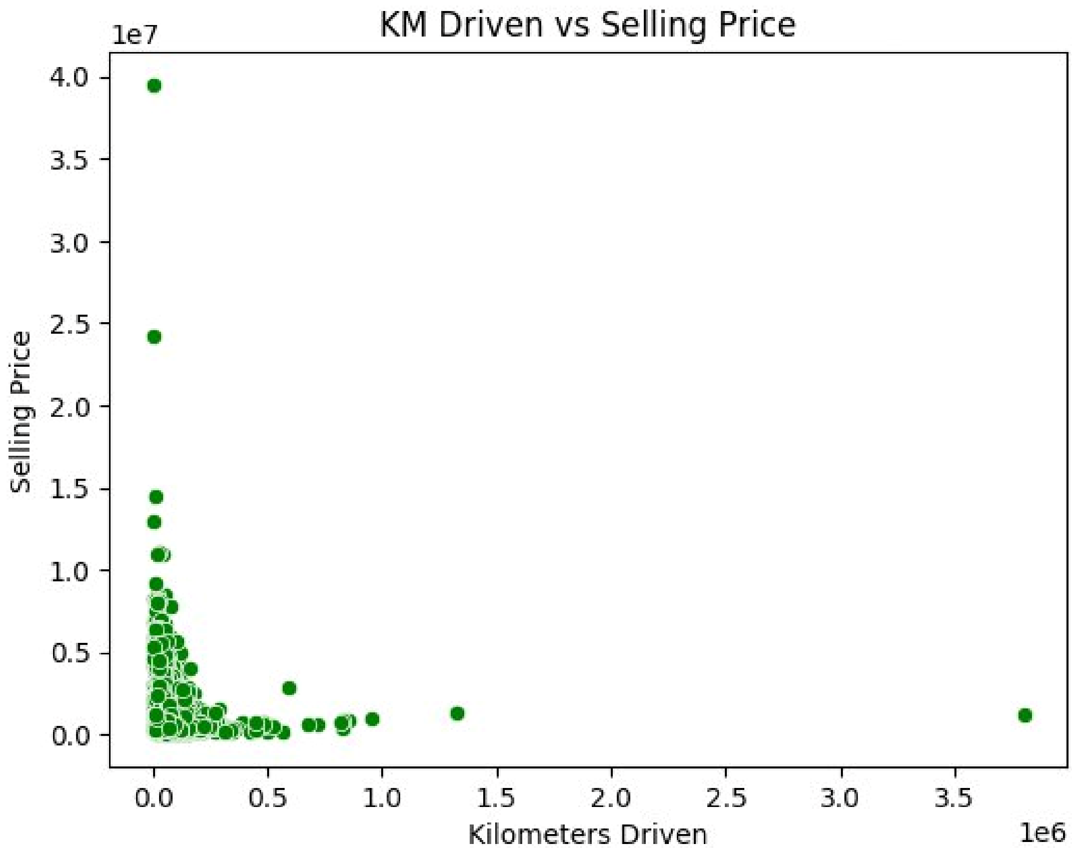
- Maruti dominates the Indian car market with ~5,000 units — nearly 67% more than second-place Hyundai (~3,000), reinforcing its position as the undisputed mass-market leader.
- Hyundai and Honda form a distant but strong second tier, suggesting a competitive mid-range segment where Korean and Japanese brands hold significant consumer trust.
- A sharp drop-off occurs after Honda (~1,500), with Mahindra, Toyota, and Ford clustering around 800–1,050 units, indicating a fragmented mid-market with no clear challenger to the top three.
- European luxury brands (BMW, Mercedes-Benz, Skoda, Audi, Volkswagen) collectively appear in the lower half, reflecting their niche premium positioning in a price-sensitive, value-driven market.
- Datsun and Jaguar's minimal presence signals either weak brand traction or limited model offerings, highlighting the difficulty of sustaining market share at both the budget and ultra-luxury ends of the spectrum.

Q2 How does vehicle age impact the selling price?

Key Insights

- A strong negative correlation exists between vehicle age and selling price, confirming the expected depreciation trend — newer vehicles (0–5 years) command significantly higher prices than older ones.
- The most extreme price outliers (~₹40M and ~₹25M) appear in the 1–4 year age range, likely representing luxury or high-end vehicles that retain premium valuations even as used cars.
- The highest price density and widest spread occur between 0–10 years, suggesting this is the most active and competitive segment of the used car market.
- Beyond 10 years, selling prices compress sharply and cluster near zero, indicating rapid value erosion and limited buyer willingness to pay a premium for aging vehicles.
- A few outliers exist at 25–29 years with near-zero prices, likely representing vintage or end-of-life vehicles with negligible resale value, pulling the tail of the distribution further right.

Q3 Does higher mileage driven reduce the selling price?

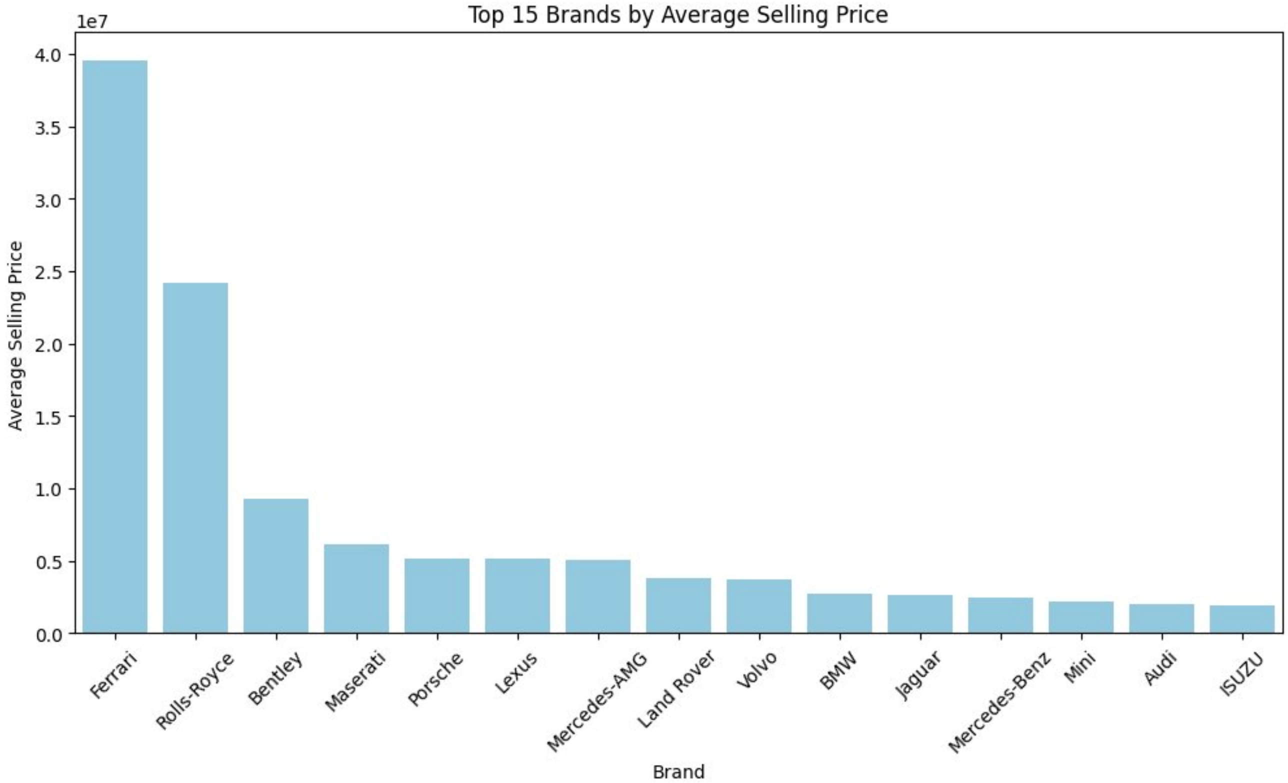


Key Insights

- A clear inverse relationship exists between kilometers driven and selling price — vehicles with lower mileage consistently fetch higher prices, validating mileage as a key depreciation driver in used car valuation.
- The highest-priced vehicles (₹25M–₹40M) are concentrated below 50,000 KM, suggesting that low-mileage luxury cars retain their premium value far better than high-usage counterparts.
- The majority of data points are densely clustered near 0–200,000 KM, indicating that most used car transactions involve relatively low-to-moderate mileage vehicles, reflecting a younger, well-maintained inventory.
- Beyond 500,000 KM, selling prices flatten near zero with very few transactions, implying that high-mileage vehicles have minimal resale appeal and represent the weakest segment of the market.

- A notable outlier at ~3.8M KM with a non-zero price likely represents a data anomaly or a specialized commercial vehicle, warranting further investigation before inclusion in predictive modeling.

Q4 Which brands have the highest average selling prices?

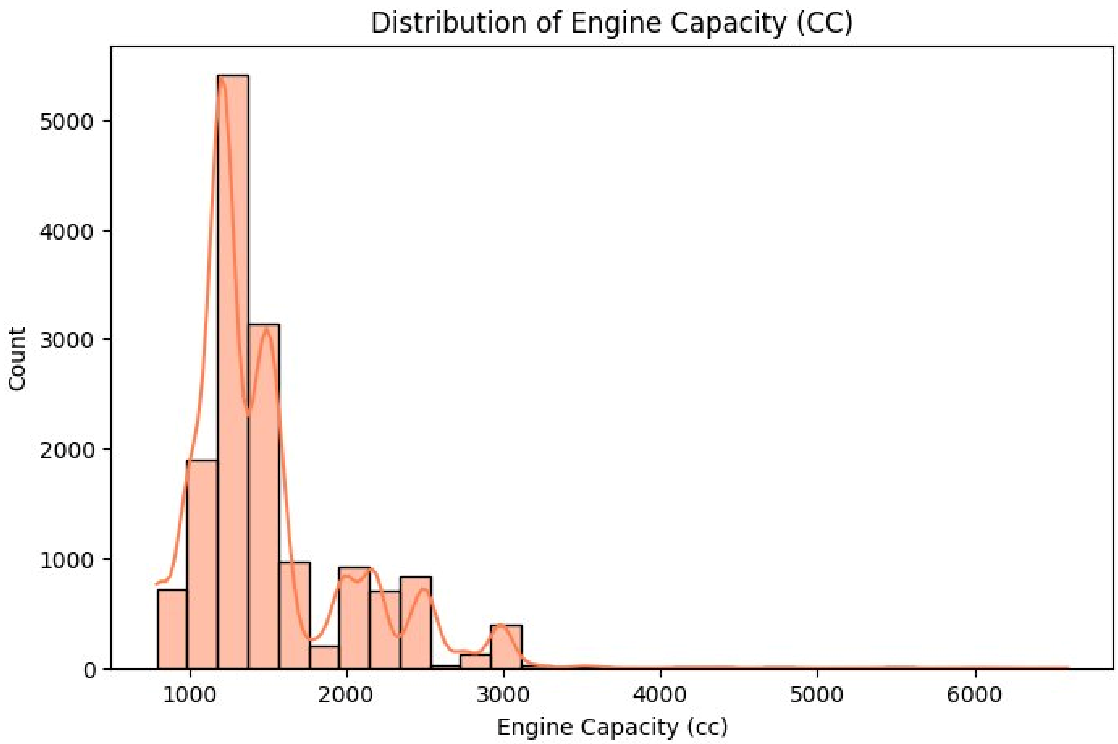


Key Insights

- Ferrari leads by a massive margin at ~₹40M average selling price, nearly 60% higher than second-place Rolls-Royce (~₹24M), cementing its status as the most aspirational and highest-valued brand in the used luxury car market.
- A steep three-tier hierarchy is visible — ultra-luxury (Ferrari, Rolls-Royce), premium luxury (Bentley, Maserati, Porsche, Lexus), and accessible luxury (BMW, Jaguar, Mercedes-Benz, Audi) — reflecting distinct consumer segments with sharply different price tolerances.
- Bentley at ~₹9.5M sits significantly below Rolls-Royce despite being in the same ultra-luxury space, suggesting either higher depreciation rates or a broader model range skewing its average downward.
- Mercedes-AMG and Land Rover cluster closely around ₹4M–₹5M, indicating strong but competitive resale positioning within the performance and SUV luxury segments respectively.

- The surprise inclusion of ISUZU in the top 15 — typically a commercial vehicle brand — hints at a niche high-value model or limited dataset entries inflating its average, flagging it as a potential outlier for data quality review.

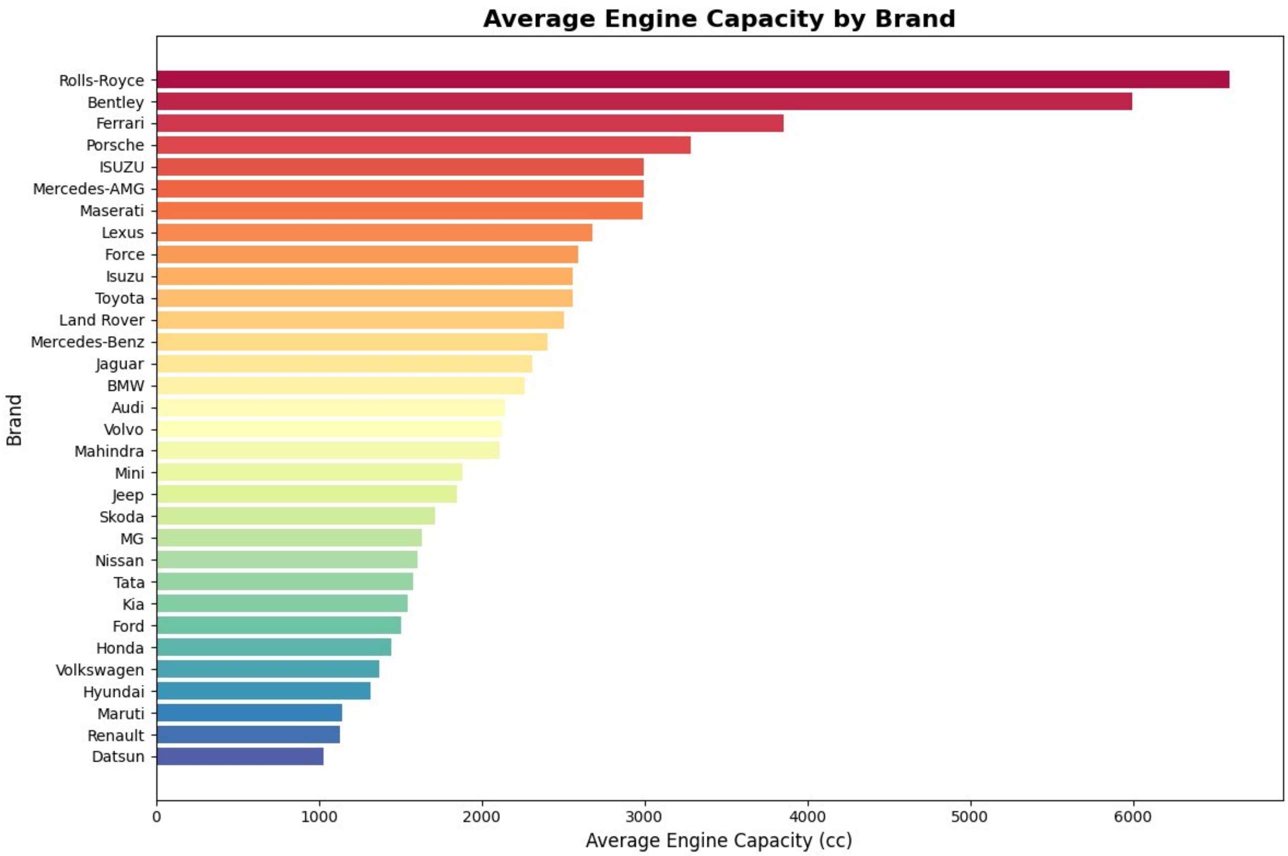
Q5 How is engine capacity distributed across the dataset?



Key Insights

- The distribution is heavily right-skewed with a dominant peak at ~1,200 CC (~5,400 units), confirming that small-engine, fuel-efficient vehicles make up the overwhelming majority of the used car market — consistent with India's price and mileage-sensitive consumer base.
- A secondary cluster is visible between 1,500–2,500 CC, representing mid-range sedans, SUVs, and diesel vehicles, forming a distinct but much smaller second tier of market preference.
- The sharp drop-off after 1,200 CC and again after 2,500 CC suggests two natural market breakpoints — one separating economy from mid-range, and another separating mid-range from premium engine segments.
- Engine capacities above 3,000 CC are extremely rare in this dataset, reflecting the high ownership cost, taxation, and limited demand for large-displacement vehicles in the Indian used car ecosystem.
- The multimodal nature of the distribution (multiple small peaks at 1,000, 1,500, 2,000, and 2,500 CC) indicates standardized engine sizing across manufacturers, with consumers gravitating toward specific, well-known engine configurations.

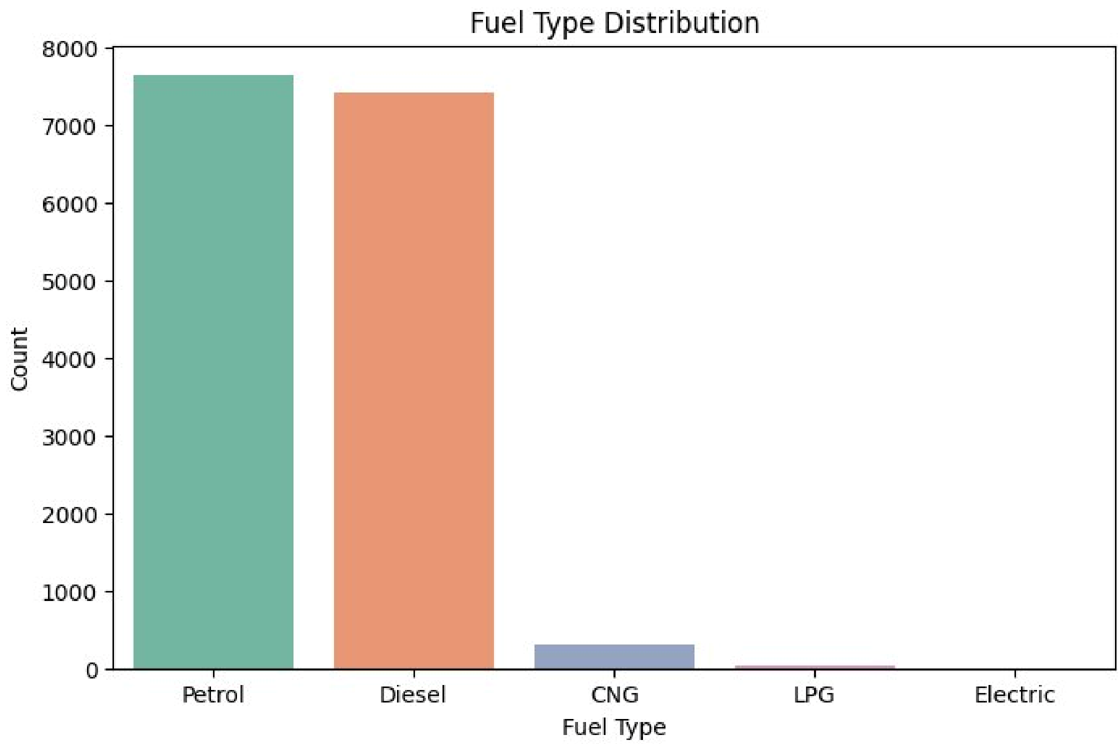
Q6 Which brands have the highest or lowest average engine capacity?



Key Insights

- Rolls-Royce (~6,500 CC) and Bentley (~6,000 CC) lead by a significant margin, reflecting their signature large-displacement, high-torque engines that are central to their ultra-luxury brand identity and driving experience.
- A clear engine-size hierarchy mirrors the pricing hierarchy — ultra-luxury brands (Rolls-Royce, Bentley, Ferrari) average 4,000–6,500 CC, while mass-market brands (Maruti, Renault, Datsun) average just 1,000–1,200 CC, confirming engine capacity as a strong proxy for brand positioning.
- The 2,500–3,200 CC mid-zone is occupied by performance and near-luxury brands like Porsche, Mercedes-AMG, ISUZU, and Maserati, forming a distinct performance-oriented cluster between mass-market and ultra-luxury segments.
- Indian domestic brands — Maruti, Tata, Mahindra, and Renault — all fall below 2,200 CC on average, consistent with their focus on affordable, fuel-efficient vehicles tailored to cost-conscious Indian consumers.
- Datsun records the lowest average engine capacity (~1,050 CC), reinforcing its positioning as an entry-level budget brand with minimal powertrain ambition, sitting at the opposite end of the spectrum from Rolls-Royce.

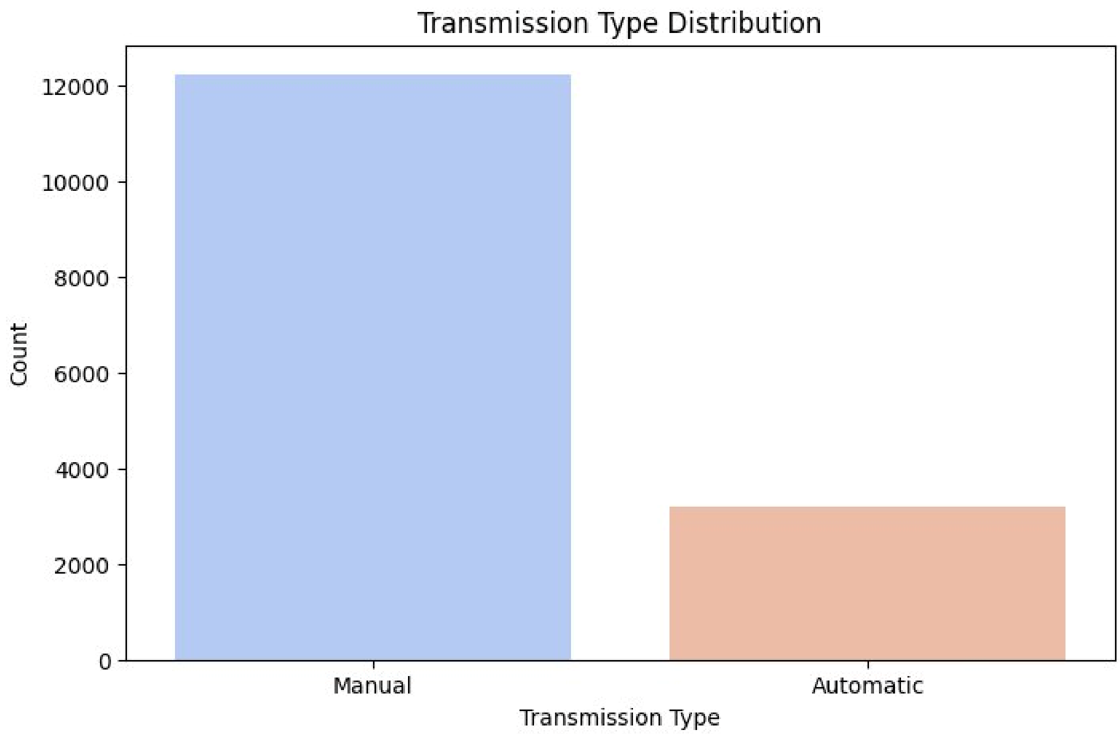
Q7 Which fuel type is most commonly used in cars?



Key Insights

- Petrol (~7,700) and Diesel (~7,500) dominate the used car market almost equally, together accounting for over 95% of all listings — reflecting the long-standing duopoly of internal combustion engines in India's automotive landscape.
- The near-parity between Petrol and Diesel suggests a balanced consumer preference, with Petrol marginally ahead, possibly due to recent diesel taxation policies and urban restrictions discouraging diesel vehicle ownership.
- CNG vehicles (~350) represent a distant third, indicating a slowly growing but still niche alternative fuel segment — primarily driven by urban commuters seeking lower running costs amid rising petrol and diesel prices.
- LPG and Electric vehicles are nearly invisible in the dataset, collectively accounting for less than 1% of listings, highlighting the nascent stage of alternative fuel adoption in India's used car ecosystem.
- The near-zero Electric vehicle presence is particularly telling — despite growing new EV sales, the used EV market remains virtually non-existent, suggesting low resale activity, consumer hesitation around used battery health, or limited early EV ownership.

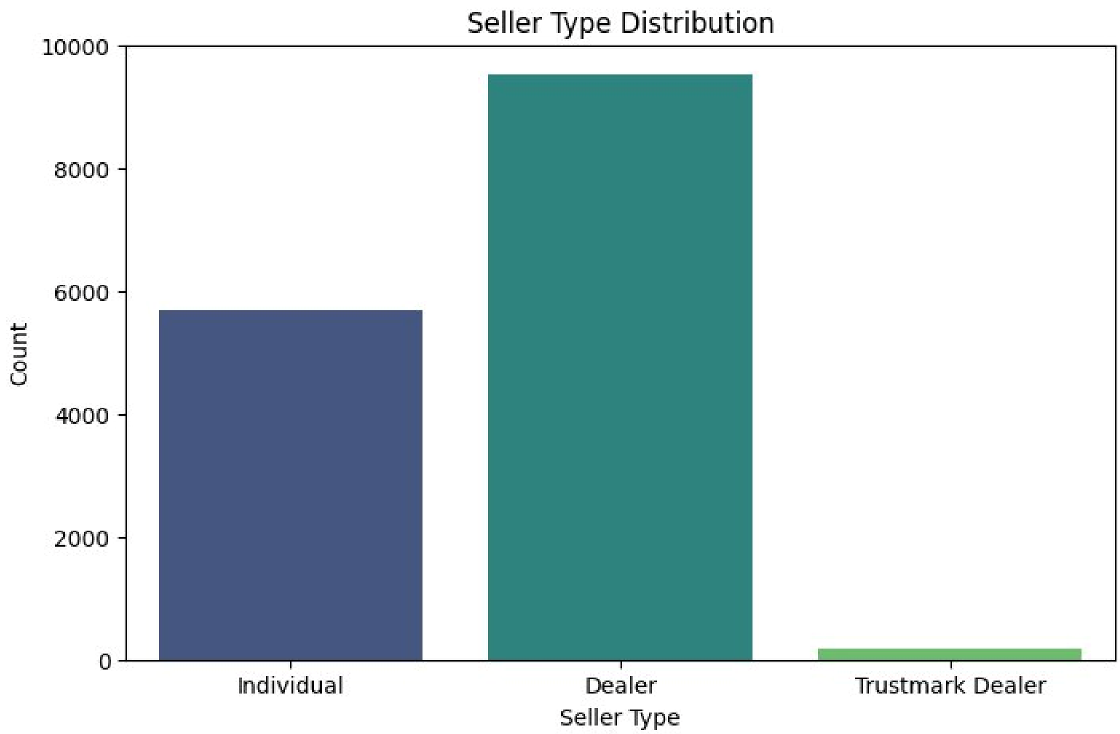
Q8 What is the distribution of transmission types?



Key Insights

- Manual transmission overwhelmingly dominates the used car market at ~12,200 units versus ~3,200 Automatic, a roughly 80:20 split — reflecting India's deeply entrenched preference for manual gearboxes driven by lower purchase cost and cheaper maintenance.
- The 4:1 ratio between Manual and Automatic listings suggests that automatic vehicles, while growing in new car sales, have yet to make a significant dent in the used car supply chain, likely due to their relatively recent mass-market adoption in India.
- The limited Automatic inventory (~3,200 units) could create a supply-demand imbalance, potentially driving higher resale prices for automatic vehicles as urban buyer preference gradually shifts toward convenience-oriented driving.
- The dominance of manual transmissions aligns strongly with the earlier finding of small-engine, budget-friendly vehicles (Maruti, Hyundai, Honda) leading the market — segments where manual gearboxes remain the default offering.
- As India's urban infrastructure grows and traffic congestion worsens, the Automatic segment is expected to grow significantly in future datasets, making current automatic listings a potentially undervalued and high-demand asset class.

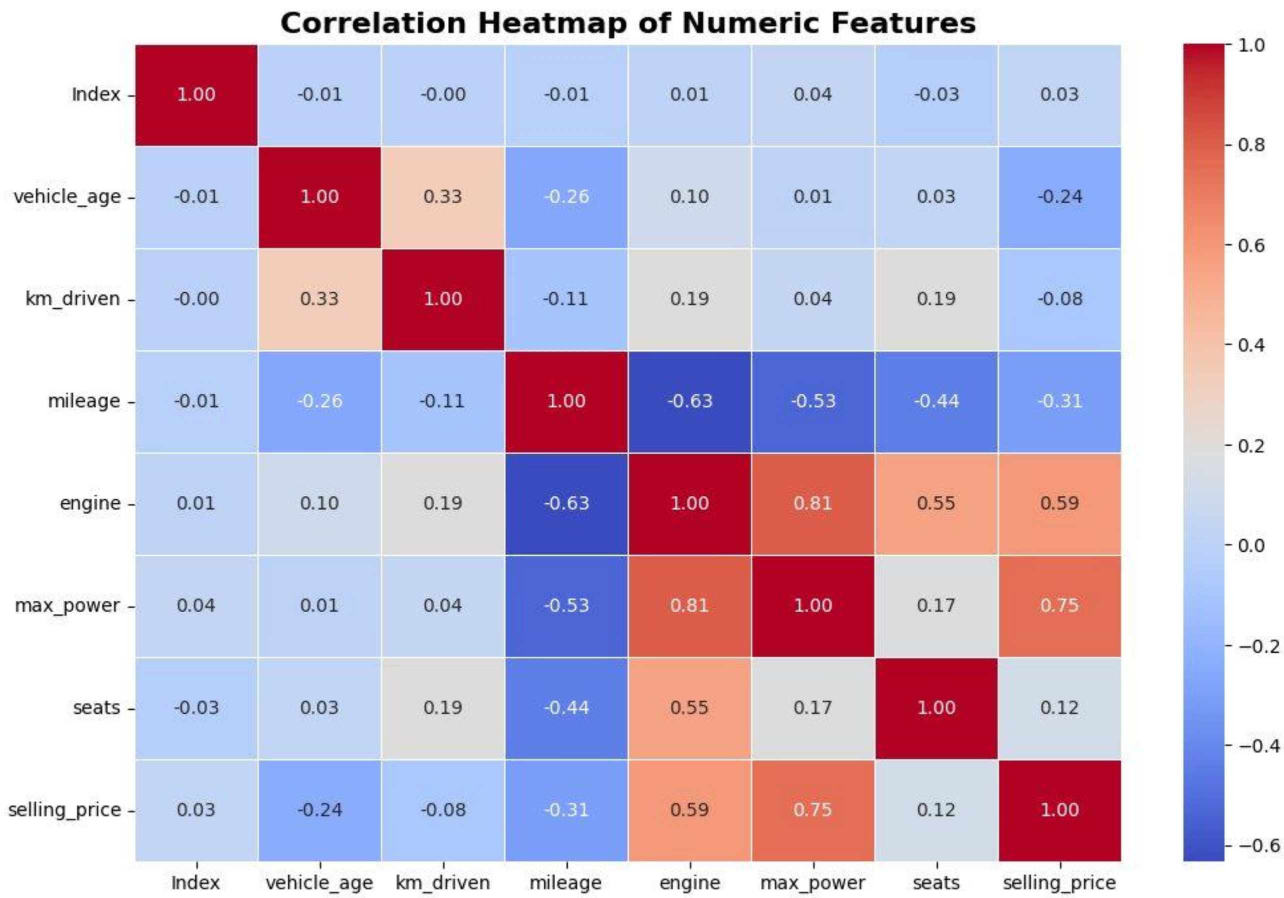
Q9 Which seller type dominates the used car market?



Key Insights

- Dealers dominate the used car market at ~9,500 listings (~62%), nearly double the Individual sellers (~5,700 at ~37%), indicating a highly organized and commercially driven resale ecosystem rather than a peer-to-peer marketplace.
- The strong dealer presence suggests increasing professionalization of India's used car industry, with organized players like CarDekho, Cars24, and OLX Autos likely contributing significantly to the dealer-listed inventory.
- Individual sellers at ~37% still represent a substantial share, indicating that private peer-to-peer transactions remain a significant and trusted channel — often preferred by buyers seeking negotiation flexibility and direct ownership history.
- Trustmark Dealers account for a negligible ~150 listings (<1%), suggesting that certified or quality-assured dealer programs are still in their infancy in India's used car market, representing a significant untapped growth opportunity.
- The near-absence of Trustmark Dealers points to a trust gap in the market — as consumer awareness around vehicle quality certification grows, this segment could expand rapidly, reshaping buyer confidence and pricing premiums.

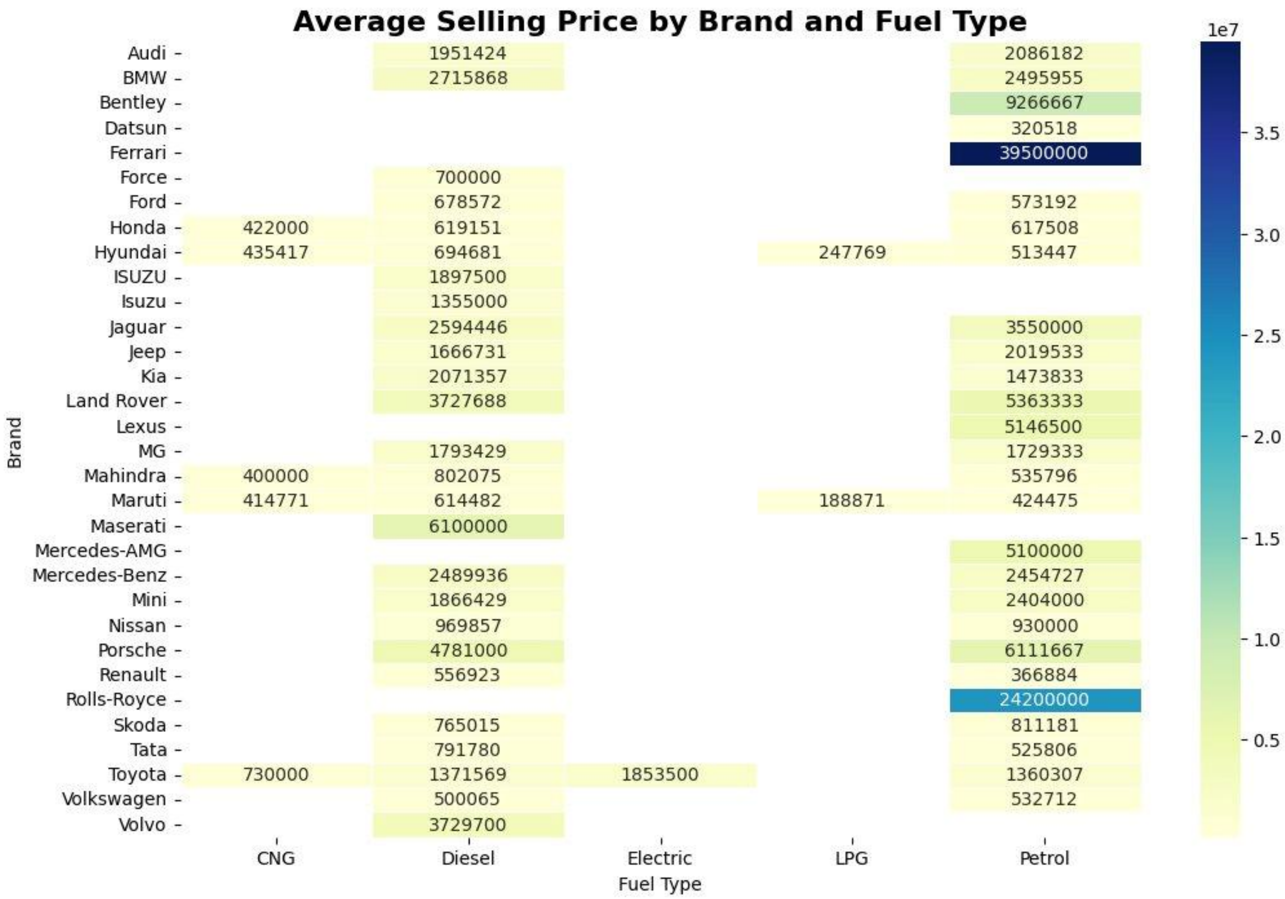
Q10 Are there any strong correlations between numeric features?



Key Insights

- Max power (0.75) and engine capacity (0.59) are the strongest positive predictors of selling price, confirming that performance-oriented specifications drive premium valuations far more than age or mileage in the used car market.
- Mileage shows a strong negative correlation with both engine size (-0.63) and max power (-0.53), revealing a fundamental trade-off — high-performance, large-engine vehicles are inherently less fuel-efficient, creating two distinct buyer segments: economy-focused vs performance-focused.
- Vehicle age (-0.24) and km driven (-0.08) have surprisingly weak correlations with selling price, suggesting that engine power and capacity are far more dominant pricing factors than depreciation-based metrics — a critical insight for building accurate price prediction models.
- The strong engine-to-max_power correlation (0.81) indicates near-multicollinearity between these two features, meaning only one may be needed in a regression model to avoid redundancy and overfitting.
- Mileage's negative correlation with selling price (-0.31) and seats (-0.44) suggests that fuel-efficient, smaller-engine vehicles tend to have fewer seats, reinforcing the pattern that compact economy cars dominate the low-price, high-mileage segment of the market.

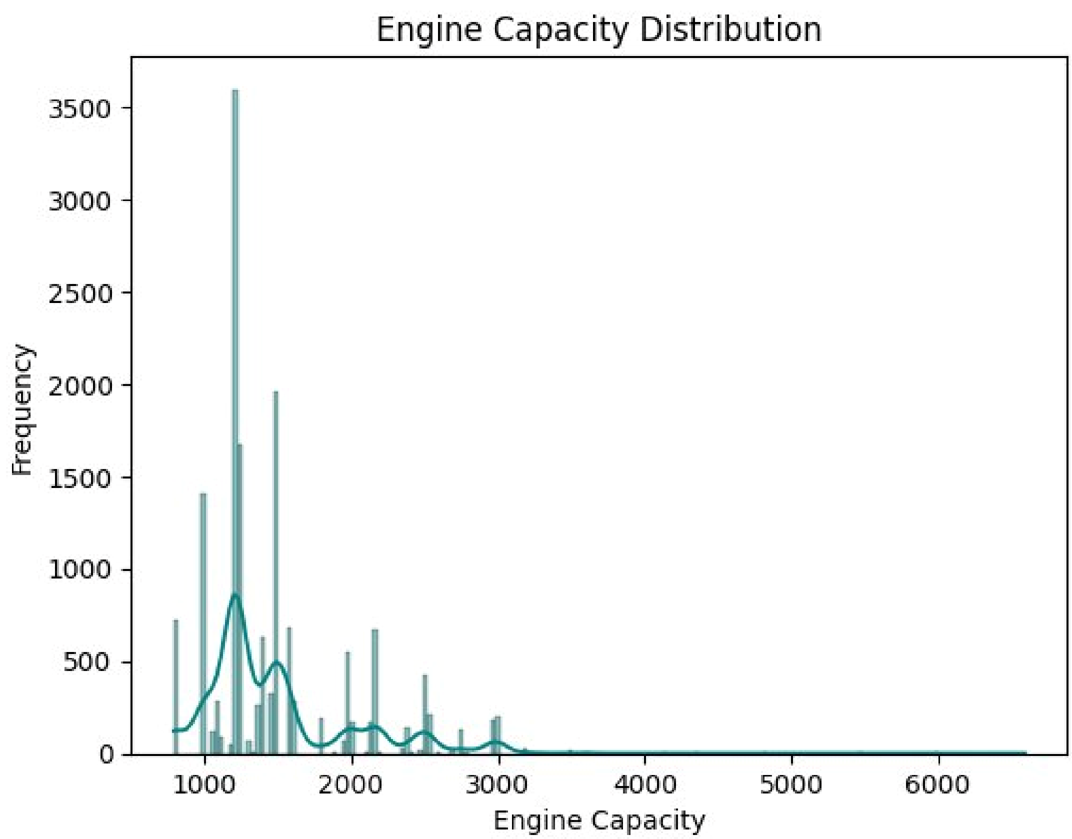
Q11 How does average selling price vary by brand and fuel type?



Key Insights

- Ferrari Petrol (~₹39.5M) and Rolls-Royce Petrol (~₹24.2M) stand out as extreme high-value outliers — both exclusively petrol-powered, reinforcing that ultra-luxury brands have zero dependency on alternative fuels and command unmatched resale premiums.
- Maserati Diesel (~₹6.1M) and Porsche Diesel (~₹4.78M) are the highest diesel-priced entries, indicating that luxury diesel variants retain strong value due to their performance-tuned engines appealing to a niche but affluent buyer segment.
- Mass-market brands like Maruti, Hyundai, and Honda show multi-fuel presence (CNG, Diesel, Petrol, LPG), with prices tightly clustered between ₹188K–₹694K — reflecting their broad, affordable product range designed to cater to every fuel preference and budget.
- Toyota is the only brand with a notable Electric average (~₹1.85M), highlighting its early mover advantage in the hybrid/electric space, while the near-complete absence of Electric entries across other brands confirms the used EV market remains largely undeveloped.
- LPG presence is limited to only Hyundai (~₹247K) and Maruti (~₹188K), both at very low price points — suggesting LPG is a budget-driven aftermarket modification losing relevance as CNG infrastructure expands across Indian cities.

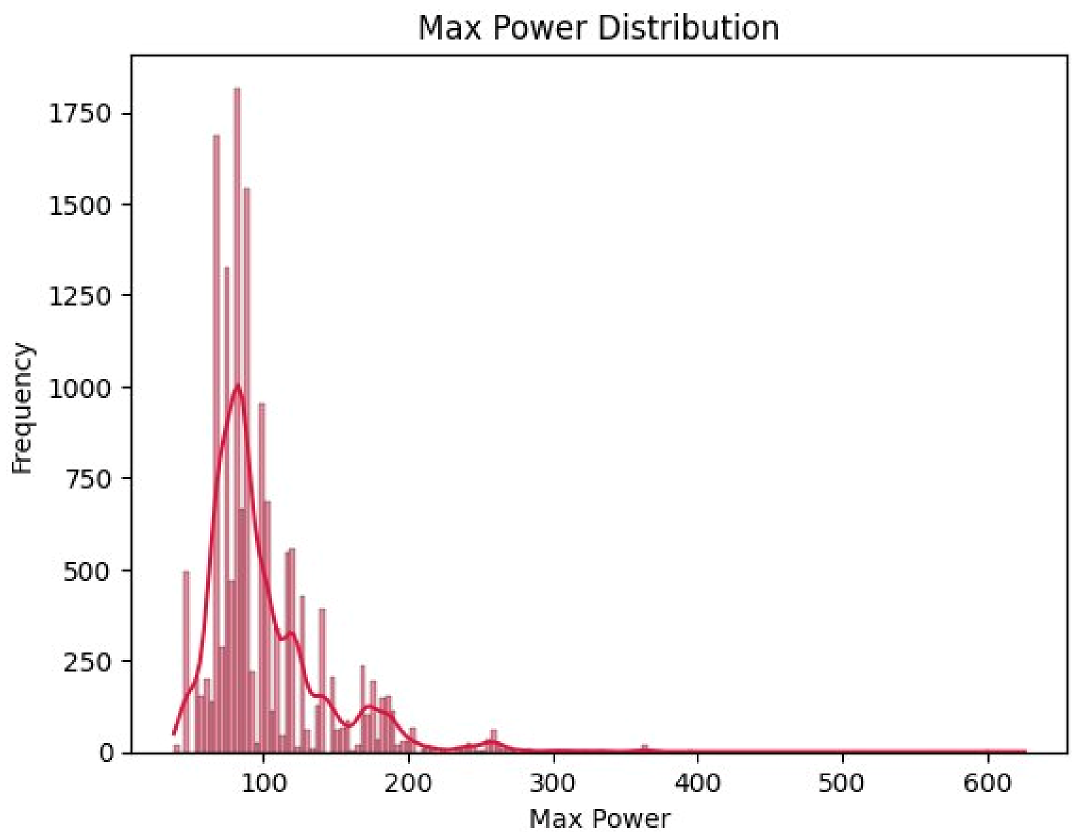
Q12 How are engine capacities distributed ?



Key Insights

- The distribution is sharply right-skewed with a dominant spike at ~1,200 CC (~3,600 frequency), confirming that small-displacement engines are by far the most common in the used car market — driven by India's strong preference for fuel-efficient, affordable vehicles.
- Multiple distinct spikes are visible at standardized engine sizes (998, 1197, 1248, 1498, 1968, 2494 CC), revealing that manufacturers deliberately cluster around specific displacement milestones — likely to optimize tax brackets, emission norms, and segment positioning.
- A secondary cluster of spikes between 1,500–2,500 CC represents the mid-range diesel and SUV segment, with notably lower frequencies (~200–550) — confirming these are niche but consistent choices among buyers seeking more power and load capacity.
- Engine capacities beyond 3,000 CC are extremely sparse, with near-zero frequency across the 3,000–6,700 CC range — isolating large-displacement vehicles as a rare luxury segment with negligible volume contribution to the overall used car pool.
- The highly granular, spiky nature of the distribution underscores that engine capacity is a discrete, manufacturer-defined specification — an important consideration when engineering features for machine learning price prediction models.

Q13 What is the distribution of maximum power values?



Key Insights

- The distribution is heavily right-skewed with a dominant cluster between 60–100 bhp (~1,800 peak frequency), confirming that low-to-moderate power output vehicles make up the vast majority of used car listings — consistent with the dominance of small-engine, budget-friendly brands like Maruti and Hyundai.
- Multiple sharp spikes between 60–150 bhp mirror the engine capacity distribution's spiky pattern, reinforcing that max power is a manufacturer-defined discrete specification tightly coupled to specific engine configurations rather than a continuously varying attribute.
- A secondary, flatter cluster between 150–250 bhp represents performance-oriented mid-range vehicles (SUVs, diesel sedans, entry-level luxury cars), forming a distinct but low-volume power segment with moderate market presence.
- Power outputs beyond 300 bhp are extremely rare, appearing as near-zero frequency outliers — isolating high-performance luxury and sports cars (Ferrari, Rolls-Royce, Porsche) as a negligible volume but high-value niche within the dataset.
- The strong alignment between this distribution and the engine capacity distribution further validates max power and engine CC as highly correlated features (r = 0.81), reinforcing the case for using only one in a lean predictive pricing model.

Outlier Detection & Statistical Analysis

Outlier Detection Methods

Two complementary methods were applied to detect extreme outliers in the selling_price distribution. Both methods consistently identify the same ultra-luxury brands as high-end outliers.

These entries should be carefully treated during ML model preprocessing.

Method	Approach	Treatment Recommendation
Z-Score	Flag records with $\lvert z \rvert > 3$ standard deviations from the mean price	Log-transform selling_price; cap or segment luxury brands above ₹10M
IQR	Flag records outside Q1 − 1.5×IQR to Q3 + 1.5×IQR price range	Remove or Winsorize extreme values; train separate model for luxury segment

Statistical Properties of Selling Price

The selling_price distribution exhibits strong positive skewness and high kurtosis, driven by the extreme price points of ultra-luxury vehicles. These statistical properties have important implications for machine learning model design and feature preprocessing.

Statistic	Observed Value	Implication for Modeling
Skewness	High Positive	Majority of cars are low-priced; a few extreme luxury outliers pull the mean upward
Kurtosis	High (Leptokurtic)	Heavy-tailed distribution; outliers exert disproportionate statistical influence
Distribution Shape	Right-skewed	Log transformation of selling_price strongly recommended before model training
Outlier Brands	Ferrari, Rolls-Royce, Bentley	Consider segmenting luxury vs mass-market models or applying log/cap treatment

Key Takeaways

The following table consolidates the most critical findings from this EDA and their direct implications for both business intelligence and machine learning model development.

#	Finding	Implication
1	**Maruti dominates**	Most listed brand — India's undisputed mass-market used car choice
2	**Max Power (r=0.75)**	Strongest predictor of selling price — performance specs matter most
3	**Engine CC (r=0.59)**	Second strongest predictor; highly collinear with Max Power (r=0.81)
4	**Vehicle Age (r=-0.24)**	Weak price predictor — performance specs outweigh depreciation factors
5	**Petrol + Diesel = 95%+**	ICE duopoly dominates — EV and LPG adoption negligible in used market
6	**Manual = 80% share**	India's cost-sensitive buyers strongly prefer manual transmission
7	**Dealers = 62% listings**	Used car market increasingly professional and organized
8	**Right-skewed prices**	Log transformation of selling_price strongly recommended before modeling

Next Steps & Recommendations

Based on the patterns and findings revealed in this EDA, the following pipeline is recommended for building a production-grade used car price prediction model:

Step	Phase	Action
1	**Data Cleaning**	Handle missing values in mileage, engine, max_power, and seats columns
2	**Outlier Treatment**	Cap or remove price outliers using IQR / Z-score; segment luxury brands
3	**Feature Engineering**	Log-transform selling_price; encode categorical features (brand, fuel_type, etc.)
4	**Baseline Modeling**	Train Linear Regression and Ridge/Lasso as baseline models
5	**Advanced Models**	Build Random Forest, XGBoost, and LightGBM for higher accuracy
6	**Feature Importance**	Analyze top predictors; confirm max_power vs engine redundancy
7	**Model Evaluation**	Compare RMSE, MAE, and R^2 across models with cross-validation
8	**Deployment**	Build a Flask/Streamlit app for real-time used car price prediction

Conclusion

This Exploratory Data Analysis of the CarDekho Used Car Dataset provides a comprehensive picture of India's used car market landscape. The market is structurally bifurcated between a high-volume, budget-driven mass-market tier and a low-volume, high-value luxury tier — each with distinct pricing dynamics and buyer behavior patterns.

The analysis establishes that maximum power output and engine capacity are the primary determinants of used car resale value, outweighing traditional depreciation factors like vehicle age and kilometers driven. Petrol and diesel vehicles continue to dominate the market by an overwhelming margin, while electric vehicle adoption remains negligible in the used car ecosystem.

The used car supply chain is increasingly professionalized, with dealers accounting for nearly two-thirds of all listings — a trend that mirrors the formalization of India's organized retail sector. Trustmark certification represents a significant untapped growth opportunity that could reshape buyer trust and price premiums in the coming years.

These findings provide a robust analytical foundation for subsequent machine learning price prediction modeling, with clear guidance on feature selection, outlier treatment, and preprocessing transformations necessary to build accurate, generalizable models.

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Tools: Python · Pandas · NumPy · Matplotlib · Seaborn · SciPy